

SALMAN ALHARBI

Decision Intelligence Specialist | Data Science & MLOps

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SUMMARY

Decision Intelligence & Data Science Specialist with practical experience delivering applied AI solutions, data automation, and technical training across leading Saudi entities, including Tuwaiq Academy (with MoE), the Saudi Electricity Regulatory Authority (SERA), and the National Center for Meteorology (NCM). Experienced in deploying Local LLMs to maintain full data privacy, streamlining large-scale data workflows, and aligning processes with NDMO and Saudi PDPL regulations to support effective decision-making.

EXPERIENCE

Tuwaiq Academy (in partnership with MoE) – Jeddah, Saudi Arabia Lead Technical Instructor | High School Track (Ages 15–17) 07/2026 – 08/2026

- Cohort Leadership:** Led high school tracks across male and female cohorts, coordinating daily operations and acting as primary liaison with academy leadership.
- Deep-Tech Delivery:** Instructed university-grade modules in Generative AI (Prompt Engineering) and Cybersecurity (adversarial threat simulations).
- Applied Prototyping:** Mentored students in building IoT systems using Arduino sensors and interactive VR environments
- Event & Talent Direction:** Orchestrated the final tech exhibition and coached student speakers for stage delivery, earning executive leadership commendations.

Saudi Electricity Regulatory Authority (via UNIVERSAL STEPS GROUP) – Riyadh, Saudi Arabia 12/2025 –6/2026
Data Science Specialist Tamheer

- Sourcing Innovation:** Engineered an ingestion pilot harvesting a **4-year data backlog (500K+ records) in 15 mins** for a primary regulator (SERA); scaled the framework **to support 4 additional enterprise clients**.
- Process Optimization:** Compressed the Awareness-to-Analysis decision cycle from hours to <5 seconds using **local LLMs** to guarantee **100% data residency per PDPL standards**.
- Decision Intelligence:** Automated operational bottlenecks and transformed multi-source data into executive-ready insights for strategic regulatory oversight.
- Sentiment Analytics:** Analyzed 5+ platforms to ensure high-fidelity institutional reputation and responsiveness.

National Center of Meteorology (NCM) – Jeddah, Saudi Arabia 06/2024 – 08/2024

Data Engineer Intern

- Automated **hourly weather report processing** using **Python (Pandas, Regex)**, eliminating **manual entry**.
- Developed **ICAO message classification system** for **flight arrivals** via **time-series analysis**.
- Optimized **SQL queries** in **DuckDB**, improving **database performance**.
- Refactored **legacy code** with **OOP and modular design** for maintainability.
- Implemented **Docker containers** and **Git** for **environment consistency and version control**.

EDUCATION

Bachelor of Data Science

College of Computer Science and Engineering | University of Jeddah 08/2020 – 01/2025

- Coursework:** Machine Learning, Data Mining, Big Data Analytics, Database Systems, Cloud Computing, NLP.

PROJECTS

Automated Insurance Decision Support System | Graduation Project 10/2024 – 01/2025

- Built **YOLOv8 & EfficientNetB0** models: **81% detection, 93% severity, 100% part ID**.
- Processed **23,000+ images** using **OpenCV/Pandas**, creating a high-quality dataset.
- Designed **rule-based cost estimator** integrating all models' outputs.
- Developed unified platform combining **detection, classification & part identification (21 parts)**.
- Deployed **Streamlit app** generating **automated PDF reports**.

Saudi Real Estate Investment Valuation Engine | Predictive Analytics 10/2023 – 11/2023

- Predicted **real estate prices** with **97.2% accuracy R-squared 0.919** using **RandomForest Regressor**.
- Built **end-to-end ML pipeline with automated feature engineering**.
- Benchmarked **7 ML models** to **optimize performance**.

CERTIFICATIONS

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| Google Data Analytics Specialization Google | 04/2023 |
| Calculus for Machine Learning and Data Science DeepLearning.AI | 10/2023 |
| Linear Algebra for Machine Learning and Data Science DeepLearning.AI | 11/2023 |
| Python University of Michigan | 05/2025 |
| Microsoft 365 Fundamentals Specialization Microsoft | 07/2025 |

SKILLS

- Technical Skills:** Python, R, SQL, SAP, Power BI, ETL, Data Pipelines, Docker, Git, Tableau, Machine Learning, Deep Learning, NLP, Computer Vision, Predictive Modeling, Time-Series Analysis, Statistical Analysis.
- Soft Skills:** Problem Solving, Analytical Thinking, Teamwork, Collaboration, Leadership, Time Management, Adaptability.
- Governance & Ethics:** Saudi PDPL, NDMO Frameworks, Data Ethics, Regulatory Compliance
- Decision Intelligence:** ROI Analysis, Data Storytelling, Market Surveillance.
- Languages:** Arabic (Native), English.